



CSE110: Programming Language I (Lab)
Summer 2024
Lab Quiz - 05
Duration: 20 Minutes

10 ✓

A

Name: Md. Tahsin Hossain	ID: [REDACTED]	Section: 05
--------------------------	----------------	-------------

Question 1 [C02] [10 Points]

Trace the following code, create a tracing table and write the outputs. Your outputs won't be accepted without the workings.

1	public class A{
2	public static void main(String[] args) {
3	int x = 3, y = 3, z = 2;
4	while (x < 11) {
5	y = x - 3;
6	while (y < 6) {
7	z = (x + y * 4) * 2 + 6 / 2;
8	System.out.println(z++);
9	y += 1;
10	}
11	if (x > 9) {
12	x = 2 % x;
13	}else{
14	x += 3 % (x + 7);
15	}
16	}
17	System.out.println(z);
18	}
19	}

Outputs
9 ✓
17 ✓
25 ✓
33 ✓
41 ✓
49 ✓
39 ✓
47 ✓
55 ✓
56 ✓

x	y	z	Output
3	3	2	9
6	0	10	17
9	1	18	25
12	2	26	33
	3	34	41
	4	42	49
	5	50	39
	6	40	47
	8	48	55
	4	56	56
	5		
	6		
	6		

$$y = 3 - 3 = 0$$

$$z = (3 + 0 \times 4) \times 2 + 6/2$$

$$= 6 + 6/2$$

$$= 9$$

$$(3 + 1 \times 4) \times 2 + 6/2$$

$$14 + 3$$

$$(3 + 8) \times 2 + 6/2$$

$$22 + 3$$

$$2+2$$

$$9 + 3 \times (9 + 7)$$

$$9+3$$

$$(3 + 12) \times 2 + 6/2$$

$$30 + 3$$

$$(3 + 16) \times 2 + 3$$

$$(3 + 20) \times 2 + 3$$

$$3 + 3 \times 10$$

$$3 + 3$$

$$6$$

$$(6 + 16) \times 2 + 3$$

$$47$$

$$13) \frac{30}{3} (0$$

$$5) \frac{30}{3} (0$$

$$10) \frac{30}{3} (0$$

$$6 + 3 \times 13$$

$$9$$

$$(6 + 12) \times 2 + 3$$